



Assigned Fair, Delivered by Favour

A Survey of Cook-Night Adherence to a New-Parent Meal-Train App's Fairness-Scored Roster

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Background

Meal-train apps compute a rotating “fairness deviation” score to decide whose turn a cook-night is. The assignment is only ever a suggestion: the app’s own logs cannot see whether it was accepted, swapped, or quietly ignored.

Aims

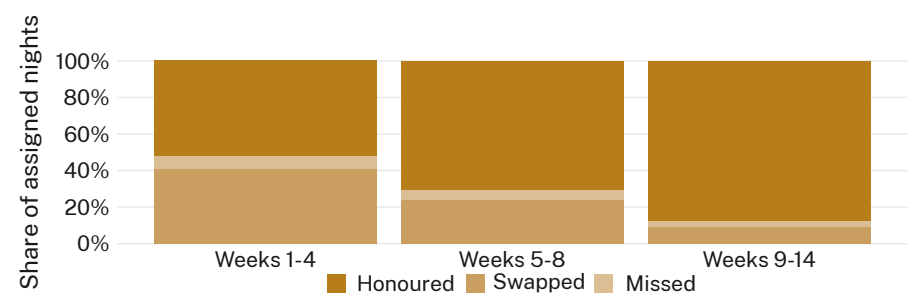
- Measure how often an app-assigned cook-night is honoured as issued versus swapped through the group’s own message thread.
- Test whether swap likelihood tracks the app’s fairness-deviation score or an informal roster-fatigue cue.
- Compare the app’s own participation tally against households’ self-reported outcomes.

Methods

- $n = 146$ households · 19 active rosters · surveyed after each of 812 assigned nights · 14 weeks.
- Mixed-effects logistic model of swap probability on fairness-deviation score, roster-fatigue flag, and week.
- Assignment log linked to self-report by night; swap reasons pre-coded, no thread content retained.

Results

The app’s fairness-deviation score showed only a weak association with swap likelihood ($r = 0.14$). A simple roster-fatigue flag — whether the assigned household had cooked in the preceding fortnight — tracked swaps far more closely ($r = 0.58$). Households who swapped almost always renegotiated directly with a neighbouring household rather than returning the night to the app for reassignment.



Swap rate fell from 41% of assigned nights in weeks one to four to 9% by weeks nine to fourteen, tracking the roster’s informal seniority order rather than the app’s fairness-deviation score.

Discussion & conclusions

The roster reads as fair on the app’s own terms while households quietly re-sequence it by a rule the app never asked about. The fairness score and the lived roster converge only once the informal order has already settled, by which point the score is confirming a pattern the group set for itself. Next: extending the audit past week fourteen, when meal-trains typically wind down, and testing whether the same roster-fatigue cue predicts handovers in other rotating-favour systems.

References

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We thank the households of the participating cook-night rosters. Self-report only; no assignment-thread message content retained; thresholds pre-registered.

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